PH2204 Mathematical Methods of Physics I
Spring Semester - 2026
Indian Institute of Science Education and Research, Kolkata
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Homework: 4 Submission Date: 12/02/2026

The hand written solutions must be submitted at the start of the class

1. Consider the following vector field $\vec{A} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$. Find out the line integration of the vector field along two parametric paths (i) $x = y = z = t$, and (ii) $x = y = z = t^2$ from $(0, 0, 0)$ to $(1, 1, 1)$. Try to visualise these paths!

2. Find work done in moving a particle once around a circle in $x - y$ plane if circle has the radius of 3, and centre at the origin. The force vector is given by $\vec{F} = (2x-y+z)\hat{i}+(x+y-z^2)\hat{j}+(3x-2y+4z)\hat{k}$. Does the result tell you something about the nature of the force? Verify the nature of the force in another way!

   If you have moved around the circle in the opposite direction, what would be the result?

3. Consider the following two-dimensional vector field

$$\vec{F} = -\frac{y}{x^2 + y^2}\hat{i} + \frac{x}{x^2 + y^2}\hat{j}.$$

   Find out $\vec{\nabla} \times \vec{F}$ explicitly. Now, calculate the line integration of the vector field around a circle of radius R, centred at origin. Are these results consistent?

4. For a conservative vector field $\vec{A} = \vec{\nabla}\phi(x, y, z)$, where ϕ is called the scalar potential associated with the field $\vec{A}$. For $\vec{A} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$, write down equations for $\frac{\partial\phi}{\partial x} = A_x$, $\frac{\partial\phi}{\partial y} = A_y$ and $\frac{\partial\phi}{\partial z} = A_z$, and solve for an unique function $\phi(x, y, z)$ which is arbitrary to the extent of adding a constant.

5. Consider the equation of motion of a particle of mass m given by

$$m\frac{d^2\vec{r}}{dt^2} = f(r)\hat{r} \ ,$$

   where $\vec{r}$ is the position vector of the particle with r being its magnitude and $\hat{r}$ is an unit vector in the radial direction.

   (a) Interpret the above equation physically for $f(r) < 0$ and $f(r) > 0$, and think about some physical systems.

   (b) Show that $\vec{r} \times \frac{d\vec{r}}{dt}$ is a constant vector, and call that $\vec{A}$

   (c) Draw a cartesian coordinate system and mark a point M with $\vec{r}$ on the x-y plane. Mark another adjacent point N where the particle has moved after sometime Δt. Now, interpret the constant vector $\vec{A}$ in part (b).

   (d) Using the above results show that the motion of the particle remains on the x-y plane.

   (e) Focussing on $f(r) = -\frac{GMm}{r^2}$ for Sun-Earth system, interpret (c) as one of Keplar's law.

6. Consider a particle of mass m moving in a conservative force field $\vec{F} = -\vec{\nabla}\phi$, and follows Newton's second law. Show that

$$\phi(A) + \frac{1}{2}mv_A^2 = \phi(B) + \frac{1}{2}mv_B^2 \ ,$$

where A and B are two arbitrary points in the space and v_A and v_B are velocities at those points. This shows conservation of energy in a conservative force field like gravitational field. Pay special attention to the negative sign in $\vec{F} = -\vec{\nabla}\phi$.

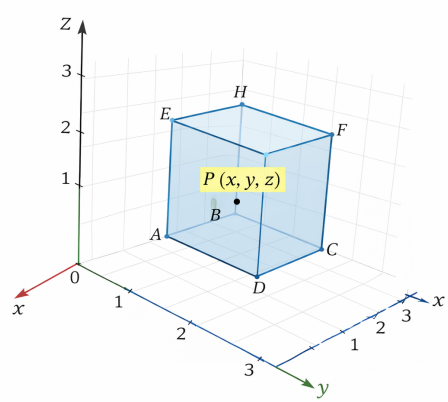
7. Evaluate $\int \vec{F} \cdot \hat{n}\ dS$ for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, where S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

8. An ellipsoid $\frac{x^2}{a^2} + \frac{x^2}{b^2} + \frac{x^2}{c^2} = 1$ encloses a fluid with density function $\rho(x, y, z) = x^2y^4$. Find the mean density. Try to make clever coordinate transformations to do integrations.

9. Consider the following cuboid as a proxy to infinitesimal volume element dV which is centred around the point $P(x, y, z)$. Also, think that the whole space is embedded with a vector field $\vec{A}$. Show that the following quantity

$$\int \vec{A} \cdot \hat{n}\ dS$$

on the following plane surfaces ADEG and BHFC is proportional to $\frac{\partial A_x}{\partial x}(x, y, z)$ where $\hat{n}$ is outward normal vectors for each of the plane surface elements dS. Do the similar exercises for opposite plane surfaces to convince yourself that

$$(\vec{\nabla} \cdot \vec{A})(x, y, z) = \lim_{\Delta V \to 0} = \frac{\oint \vec{A} \cdot \hat{n} dS}{\Delta V} \ .$$

This result should convince you that the divergence of a vector field corresponds to the net outflow of the vector quantity through the surface S per unit volume, where S is the surface that contains the volume. (You should look at notes what we did in the class for the interpretation of curl of a vector field.)